

Activity: 11) Medical Image Enhancement

i) Recommended Transformation:- Log Transformation or Power-Law (Gamma) transformation ($\gamma < 1$).

ii) Why:- Expands narrow, dark input ranges into wider output ranges, highlighting soft tissue without overexposing bright bones.

iii) Effect:- Dark soft tissues become bright and clear while bright bones remain visible.

2) Night-Time Surveillance

i) Recommend Transformation:- Log Transformation or Power Law (Gamma) transformation ($\gamma < 1$).

ii) Why it Beats Image Negative:- Inverting an image ~~$s = L - 1 - r$~~
 $\boxed{s = (L-1) - r}$ flips contrast without solving narrow dynamic range. Log / Gamma stretching specifically expands dark Pixel value (0 to 40) across the full spectrum.

iii) Appearance:- objects hidden in dark areas (pedestrians, cars) become distinct and readable.

3) Finger Print Recognition

i) Recommended Transformation:- contrast stretching (Piecewise - Linear) or Histogram Equalization.

ii) Why:- stretches Low-contrast values to span 0 to 255, creating maximum distinction b/w dark ridges and light valleys.

iii) If Log Transformation is used:- Dark ridges lighten and bright areas compress, washing out contrast and flattening the finger print detail.